



Bruton's tyrosine kinase inhibitors: a new class of multiple sclerosis therapeutics

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Bruton's tyrosine kinase (BTK) is an intracellular enzyme involved in a wide range of cellular functions. BTK inhibitors (BTKIs) are of interest in multiple sclerosis because they modulate various types of immune cells that have key roles in multiple sclerosis pathophysiological processes, including B cells and microglia. Moreover, because BTKIs are small molecules that might cross the blood–brain barrier and potentially modulate relevant cells within the CNS, there is added interest in their use in multiple sclerosis. In the past decade, there has been a flurry of drug development of BTKIs in multiple sclerosis. The BTKIs being evaluated in multiple sclerosis have differences in pharmacological properties that might result in meaningful differences in safety and efficacy. Recently reported phase 2 and phase 3 clinical trials have provided the field with renewed hope that some BTKIs could modulate progressive multiple sclerosis disease mechanisms that current disease-modifying treatments insufficiently address. However, meaningful differences in clinical outcomes and MRI findings have been reported. If the beneficial findings seen with some BTKIs are confirmed, these observations suggest the beginning of a new era of treatment in multiple sclerosis. The results of upcoming clinical trials, real-world studies, and regulatory approval decisions will guide a more nuanced approach to determining the patient populations that will benefit most from this treatment class.

Introduction

Bruton's tyrosine kinase (BTK), an intracellular enzyme expressed predominately in haematopoietic cells, is involved in a wide range of cellular functions, including maturation, proliferation, and differentiation of haematopoietic cells (figure).¹ The discovery that mutations in the gene encoding BTK cause a genetic immunodeficiency disorder (X-linked agammaglobulinemia) made evident the important role that BTK has in B-cell maturation and proliferation.² Accordingly, BTK inhibitors (BTKIs) have been evaluated for use in B-cell malignancies over the past three decades and have received regulatory approval for this indication.^{3,4}

Because of the important pathophysiological role B cells are known to have in many autoimmune disorders, BTKIs are being evaluated in a range of conditions, including multiple sclerosis, systemic lupus erythematosus, rheumatoid arthritis, chronic spontaneous urticaria, and Sjögren's syndrome.⁵ In multiple sclerosis, there is interest in BTKIs because of their ability to modulate several immune cells that have key roles in multiple sclerosis pathophysiological processes, including B cells and microglia (figure).^{6,7} As a result, there has been a flurry of BTKI development in multiple sclerosis in the past decade, resulting in the simultaneous evaluation of six BTKIs in phase 2 and phase 3 clinical trials.

The aim of this Review is to discuss the rationale supporting the development of BTKIs in multiple sclerosis, differences in their pharmacological properties, and results from clinical trials including efficacy and liver safety concerns, and to reflect on their potential use in multiple sclerosis clinical practice.

Evidence supporting the development of BTKIs in multiple sclerosis

The keen interest in BTKIs for multiple sclerosis is due on the one hand to their known effects in decreasing the

maturation and proliferation of B cells and downstream immunological pathways that lead to the propagation of inflammation mediated by the peripheral immune system, which is primarily responsible for relapsing disease activity.^{6,8} Moreover, BTKIs are small molecules that might cross the blood–brain barrier and potentially affect B cells and pathogenic microglia within the CNS, both key components of non-relapsing progressive disease^{6–10} that most current therapies cannot meaningfully modulate^{11–14} (figure).

Evaluation of postmortem multiple sclerosis brain tissue has provided insights supporting the development of BTKIs in multiple sclerosis. In demyelinated tissue regions, BTK immunoreactivity was increased compared with normal appearing tissue,¹⁵ and multiple sclerosis brain tissue bulk mRNA and protein analyses showed substantial increases of BTK in comparison to tissue from controls with no neurological diseases.¹⁵ When white-matter lesions were evaluated in progressive multiple sclerosis cases, BTK immunoreactivity colocalised almost exclusively with myeloid cells (probably microglia) within the rims of chronic lesions.^{15,16}

In-vitro studies in human and mouse whole blood have demonstrated that BTKIs reduce proximal and distal functional responses of B cells and myeloid cells, including antigen presentation, activation, proliferation, and differentiation, as well as the secretion of proinflammatory cytokines.^{16–19} As an example, tolebrutinib inhibited a BTK-dependent transcriptional signature involved in multiple sclerosis-relevant neuroinflammatory signalling pathways across mouse microglia, human induced pluripotent stem cell (hiPSC)-derived microglia, and a complex hiPSC-derived triculture system composed of neurons, astrocytes, and microglia.¹⁵

Studies suggest that BTKIs have several effects on B cells, including limiting activation, decreasing their

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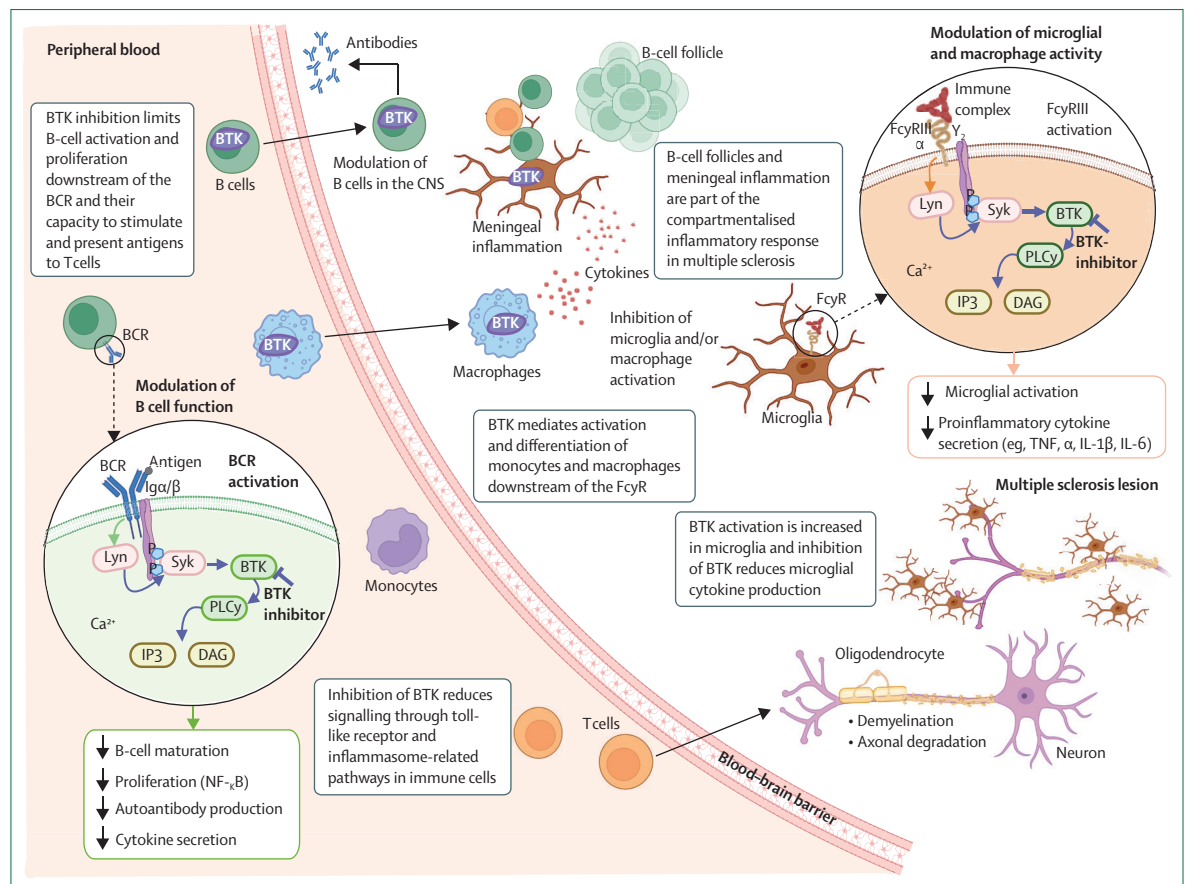


Figure: Putative mechanism of action of Bruton's tyrosine kinase in multiple sclerosis

BTK=Bruton's tyrosine kinase. BCR=B-cell receptor. FcγR=Fcγ receptor. Syk=spleen tyrosine kinase. PLCγ=phospholipase C γ. IP3=inositol triphosphate. DAG=diacylglycerol. NF=nuclear factor. TNF=tumour necrosis factor. IL=interleukin.

ability to activate T cells, and shifting the balance from proinflammatory to anti-inflammatory B-cell responses, which might be attributed to an increased proportion of transitional B cells and a decreased proportion of memory B cells. BTK inhibition shifts memory B cells away from the exaggerated oxidative phosphorylation-dependent metabolic state observed in multiple sclerosis.^{19,20}

In a range of preclinical models, BTKIs have shown functional benefit. The first of these reports dates to 2015, when the putative protein tyrosine kinase inhibitor tyrphostin AG126 was shown to directly inhibit BTK, ameliorating an experimental autoimmune encephalitis mouse model.²¹ Subsequent administration of BTKIs in both B-cell-mediated and T-cell-mediated rodent experimental autoimmune encephalitis demonstrated dose-dependent attenuation of disease-like severity.¹⁸ In B-cell-mediated experimental mouse autoimmune encephalitis, administration of evobrutinib resulted in inhibition of a range of B-cell functions, including activation, proliferation, differentiation, antigen presentation to T cells, and release of proinflammatory cytokines.^{16,18} In B-cell-mediated mouse experimental

autoimmune encephalitis, remibrutinib attenuated disease-like severity.²² In rat myelin oligodendrocyte glycoprotein-experimental autoimmune encephalitis, a preclinical model of multiple sclerosis thought to be mediated predominately by T cells and myeloid cells, remibrutinib inhibited experimental autoimmune encephalitis, and single-cell RNA profiling of brain and spinal cord tissue in these rats demonstrated reduction in neuroinflammatory pathways in microglia.²² The attenuation of disease-like severity observed in a variety of experimental autoimmune encephalitis models suggests that the beneficial effects of BTKIs extend beyond their effects on B cells and myeloid cells, influencing T-cell responses indirectly.

The effects of BTK inhibition on CNS-compartmentalised processes have also been evaluated in preclinical models. In a mouse model of chronic experimental autoimmune encephalitis, in the setting of CNS-compartmentalised inflammation, administering evobrutinib decreased proinflammatory microglial activation and enhanced phagocytic capacity, but there was no observed functional effect.¹⁶ Evobrutinib treatment showed a significant decrease in the number

of foci of leptomeningeal enhancement observed with ultra-high-field MRI, potentially a reflection of meningeal inflammation associated with cortical pathology, in comparison with vehicle control treatment in experimental autoimmune encephalitis mice.²³ In a cuprizone-induced mouse model of toxic demyelination and remyelination, evobrutinib promoted microglia-facilitated myelin debris clearance and enhanced remyelination.¹⁶ Tolebrutinib attenuated disease-like severity in experimental autoimmune encephalitis mice and decreased plasma neurofilament heavy chain concentrations, although treatment with non-CNS-penetrant BTKIs (acalabrutinib and ibrutinib) did not.¹⁵ Furthermore, analysis of single-cell RNA profiling of brain and spinal cord tissue demonstrated that BTK inhibition modulates microglial transcriptional phenotypes.¹⁵

Although there is no direct evidence supporting a role of BTK inhibition in Epstein-Barr virus (EBV)-related multiple sclerosis disease processes, such a role is plausible. The EBV latency protein LMP2A can mimic B-cell receptor signalling, with downstream signalling through BTK promoting B-cell survival (and hence viral persistence).²⁴ In turn, inhibiting BTK can kill EBV-infected B cells, as was shown with ibrutinib in the context of EBV-related lymphomas.^{3,4}

Randomised clinical trials of BTKIs in multiple sclerosis

Phase 2 clinical trials and open-label extensions

Evobrutinib was the first BTKI assessed in multiple sclerosis. In a phase 2 trial of people with relapsing multiple sclerosis, evobrutinib was evaluated at three doses.²⁵ During weeks 12–24, evobrutinib demonstrated a decrease in the primary endpoint of T1 gadolinium-enhancing (T1-Gd+) lesions in the 75 mg once-per-day group versus placebo (table 1). Secondary endpoints, including the unadjusted annualised relapse rate at week 24 and expanded disability status scale change over 24 weeks, were not statistically different across study groups. Over 52 weeks, no cases of Hy's law (liver enzyme elevations more than three times the upper limit of normal [ULN] with concurrent bilirubin more than two times the ULN) were observed (table 1). After 48 weeks, consenting participants entered the open-label extension study. No new safety signals emerged, new T1-Gd+ lesions remained low, the low annualised relapse rate observed in the phase 2 trial was sustained, and EDSS scores remained stable (table 1).²⁶ Exploratory post-hoc analyses in the phase 2 evobrutinib trial provided insights into evobrutinib's effects on CNS-compartmentalised disease processes in multiple sclerosis. The volumes of MRI-detected slowly expanding lesions evaluated over 48 weeks—hypothesised to be a marker of chronic active lesions—were lower in those treated with the highest dose of evobrutinib (75 mg twice per day) in comparison

with those treated with placebo in the first 24 weeks, then put on evobrutinib 25 mg per day in the subsequent 24 weeks, suggesting that evobrutinib might have a beneficial effect on non-relapsing progressive disease activity (table 1).³³

Tolebrutinib was the second BTKI evaluated in a phase 2 trial²⁸ of people with multiple sclerosis, which used 4-week placebo run-in and run-out periods. Four doses of tolebrutinib (5 mg, 15 mg, 30 mg, and 60 mg per day) were evaluated. At 12 weeks, there was a dose-dependent reduction of T1-Gd+ lesions in comparison with the placebo period (table 1).²⁸ Consenting participants in the phase 2 study enrolled in the open-label extension study,²⁹ and all eventually transitioned to tolebrutinib 60 mg per day. No new safety signals were identified, and participants maintained low annualised relapse rates, stable EDSS scores, and a steady rate of new lesion formation (table 1).²⁹ Prespecified endpoints included paramagnetic rim lesions, a pathologically validated marker of chronic white-matter inflammation, and slowly expanding lesions. In the subgroup of 29 participants who had paramagnetic rim lesions evaluated, 24 (83%) show no change in paramagnetic rim lesion counts.²⁹ The total volume of slowly expanding lesions decreased numerically over 96 weeks.²⁹ Additional follow-up reported up to 3·5 years revealed no new safety signals, and participants maintained low annualised relapse rates and stable EDSS scores (table 1).³⁴

More recently, fenebrutinib at 200 mg twice per day was evaluated in a placebo-controlled phase 2 trial of people with relapsing multiple sclerosis.³² At 12 weeks, the number of combined T1-Gd+ lesions at weeks 4, 8, and 12 was 0·077 (95% CI 0·043–0·135) in the fenebrutinib group and 0·245 (0·144–0·418) in the placebo group ($p=0\cdot002$). The effect of fenebrutinib on T1-Gd+ lesions relative to placebo was seen numerically as early as week 4, with the maximal effect observed at week 8 and sustained at week 12. The annualised relapse rate was 0 in the fenebrutinib group and 0·18 in the placebo group (table 1).³² No serious adverse events were observed, and liver enzyme elevations were observed in four participants, which resolved with fenebrutinib discontinuation in all cases. At week 48 of the open-label extension study, fenebrutinib maintained near-complete suppression of disease activity, with 99% of participants free of new T1-Gd+ lesions. The annualised relapse rate remained low at 0·04 and EDSS scores were stable,³² with similar findings reported at 96 weeks (table 1).³⁵

Finally, results from a phase 2 trial of orelabrutinib in people with relapsing multiple sclerosis presented as a conference abstract found a significant reduction in T1-Gd+ lesions at 12 weeks versus placebo with all three doses evaluated (table 2).³⁶

Phase 3 clinical trials

Evobrutinib was the first BTKI to be evaluated in phase 3 clinical trials: evolution RMS 1 and 2,²⁷ which were phase 3

Study design	Study population (with summary of key characteristics), total number, and duration	Study groups	Key results: efficacy	Key results: safety	
Evobrutinib, NCT02975349 ²⁵	Phase 2, multicentre, randomised, double-blind, placebo-controlled study with a parallel, open-label, active control group	Relapsing multiple sclerosis (n=267); 24 weeks duration; mean population age 42.4 years (SD 10.7); median disease duration 8.4 years (range 0.1–39.4); median EDSS 3.0 (range 0–6.0); 44% of patients with one relapse in previous 2 years; 66% of patients with at least two relapses in previous 2 years; and 39% of patients with T1-Gd+ lesions on baseline MRI	Evobrutinib 25 mg orally once per day; evobrutinib 75 mg orally once per day; evobrutinib 75 mg orally twice per day; placebo orally once per day; and dimethyl fumarate (open-label, 120 mg twice per day for the first week and 240 mg twice per day thereafter)	Fewer T1-Gd+ lesions in evobrutinib 75 mg once per day vs placebo (baseline ARR 0.30 [95% CI 0.14 to 0.63]; p=0.005) in weeks 12–24; trend towards fewer T1-Gd+ lesions with evobrutinib 75 mg twice per day vs placebo (baseline ARR 0.44 [95% CI 0.21–0.93]; p=0.06); ARR at week 24 was numerically lower in the higher-dose evobrutinib groups vs placebo (0.37 in the placebo group, 0.57 in the evobrutinib 25 mg once per day, 0.13 in the evobrutinib 75 mg once per day, 0.08 in the evobrutinib 75 mg twice per day); EDSS change over the study period was similar across study groups.	Highest rate of serious adverse events (7%) occurred in the group on evobrutinib 75 mg twice per day; grade 3 and 4 adverse events were most frequent with evobrutinib 75 mg once per day (13%), evobrutinib 75 mg twice per day (15%), and dimethyl fumarate (13%); the most common adverse events (≥5% in any treatment group) included nausea, diarrhoea, nasopharyngitis, upper respiratory tract infection, urinary tract infection, increased aspartate aminotransferase, increased alanine aminotransferase, increased lipase, increased creatinine, low lymphocyte count, arthralgia, headache, and flushing; the most common adverse events seen with any dose of evobrutinib were: nasopharyngitis (6–17%) and elevation of liver enzymes (6–11%), but no cases of Hy's law were observed; cases of elevated liver enzymes were asymptomatic and reversible; no deaths occurred.
Evobrutinib, NCT02975349 ²⁶	Long-term open-label extension of phase 2 trial (NCT02975349) with CSF substudy	Relapsing multiple sclerosis (n=164); entered OLE after 48 weeks in double-blind placebo; n=128 in OLE completed at least 192 weeks of treatment; mean population age 42.4 years (SD 10.7); median disease duration 8.4 years (range 0.1–39.4); median EDSS 3.0 (range 0–6.0); 44% of patients with one relapse in previous 2 years; 66% of patients with at least two relapses in previous 2 years; and 39% of patients with T1-Gd+ lesions on baseline MRI	Evobrutinib 75 mg orally once per day (based on original protocol); once the twice-per-day dose was found to be superior (after analysis of double-blind placebo and masked extension analyses), all participants transitioned to open-label evobrutinib 75 mg orally twice per day	Low ARR observed in the core phase 2 trial was sustained (0.11 [95% CI 0.04–0.25]); no change in median EDSS from baseline to week 192 regardless of double-blind placebo treatment status; new T1-Gd+ lesions remained low (mean <1 T1-Gd+ lesion at 192 weeks); T2 lesion volume remained stable across evobrutinib DBP treatment groups from baseline to week 192; Z scores of serum NFL decreased to levels expected in sex and BMI-matched healthy controls by week 144; slowly expanding lesion volumes were lower in those treated with the highest dose of evobrutinib (75 mg twice per day) compared to those treated with placebo for 24 weeks then 25 mg evobrutinib once per day (–474.5 mm ³ [95% CI –1098.0 to –3.0]; p=0.047) over 48 weeks	No new safety signals were identified from OLE initiation to week 192; the majority of TEAEs during OLE were mild (62%) or moderate (48%); no clinically remarkable elevations in liver enzymes were observed in OLE after patients switched from lower doses of evobrutinib to 75 mg twice per day, or after prolonged treatment

(Table 1 continues on next page)

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Study design	Study population (with summary of key characteristics), total number, and duration	Study groups	Key results: efficacy	Key results: safety
(Continued from previous page)				
Evobrutinib, NCT04338022 multicentre, randomised, double-blind, double-dummy, active-controlled trials	Relapsing multiple sclerosis (n=2290); minimum 24 weeks to maximum 156 weeks; mean age 37 years (SD 9); median disease duration 4.6 years (IQR 0–36.0); median EDSS 2.5 (range 0–5.5); mean number of relapses in previous year: Evolution RMS 1 with evobrutinib 1.2 (SD 0.5) and teriflunomide 1.2 (0.5) and Evolution RMS 2 with evobrutinib 1.3 (0.5) and teriflunomide 1.2 (0.5); patients with T1-Gd+ lesions on baseline MRI: Evolution RMS 1 with evobrutinib 208 (37.1%), teriflunomide 199 (35.3%) and Evolution RMS 2 with evobrutinib 221 (37.9%) and teriflunomide 214 (36.7%)	Evobrutinib 45 mg orally twice per day; and teriflunomide 14 mg orally once per day	ARR was similarly low in the evobrutinib and teriflunomide groups in evolution RMS 1 and 2 (0.15 [95% CI 0.12–0.18] with evobrutinib vs 0.14 [0.11–0.18] with teriflunomide; adjusted relative risk 1.02 [0.75–1.39], p=0.55 in evolution RMS 1; 0.11 [0.09–0.13] with evobrutinib vs 0.11 [0.09–0.13] with teriflunomide, adjusted RR 1.00 [0.74–1.35], p=0.51 in evolution RMS 2; no difference in 12-week or 24-week confirmed disability progression or 24-week confirmed disability improvement between groups (pooled analysis); adjusted mean number of T1-Gd+ lesions per scan was numerically higher for evobrutinib vs teriflunomide (0.50 [0.41–0.60] vs 0.35 [0.29–0.42] in evolution RMS 1 and 0.50 [0.42–0.59] vs 0.31 [0.26–0.27] in evolution RMS 2); T2 hyperintense lesions were not superior with evobrutinib vs teriflunomide; adjusted geometric mean serum NFL concentrations at week 12 showed no difference between evobrutinib and teriflunomide in evolution RMS 1 and a nominal difference in evolution RMS 2; weak evidence of an effect of evobrutinib on reducing SEL volume vs teriflunomide at week 96 (mean volume of SEL in evobrutinib vs teriflunomide: 1006.2 mm ³ [SD 201.1:18] vs 1183.9 [2151.31] in evolution RMS 1 and 985.6 [1649.17] vs 1201.7 [1825.14] in evolution RMS 2); progression independent of relapse activity, Symbol Digit Modalities Test, thalamic volume, whole-brain volume, and paramagnetic rim lesion count were not different between treatment groups	TEAEs proportions were similar between groups (85.6% and 87.2% with evobrutinib vs teriflunomide); the most frequent TEAEs (evobrutinib vs teriflunomide) were COVID-19 (19.6% vs 19.5%), ALT increase (15.2% vs 17.8%), and headache (15.4% vs 15.4%); serious TEAEs were more common with evobrutinib vs teriflunomide (7.5% vs 5.6%); liver enzyme elevations greater than five times the upper limit of normal were seen more commonly with participants treated with evobrutinib, particularly in the first 12 weeks (5% vs <1%); three patients who received evobrutinib met Hy's law; all were asymptomatic and normalised after discontinuing evobrutinib; two deaths occurred, neither deemed to be related to study treatment (viral respiratory tract infection in the evobrutinib group and suicide in the teriflunomide group)
Tolebrutinib, NCT03889639 ²⁸ Phase 2b, randomised, multicentre, double-blind, placebo-controlled, crossover, dose-finding trial	Relapsing multiple sclerosis (n=130); duration 16 weeks; mean population age in tolebrutinib group 37 years (SD 10) and in placebo group 36 years (10); mean disease duration with tolebrutinib 7.8 years (SD 7.4) and placebo 7.7 years (7.4); mean EDSS with tolebrutinib 2.5 (SD 1.1) and placebo 2.6 (1.1); mean number of relapses in previous year with tolebrutinib 1.2 (SD 0.6) and placebo 1.2 (0.5); 35% of patients with T1-Gd+ lesions on baseline MRI with tolebrutinib and 38% with placebo	Cohort 1: tolebrutinib orally once per day (one of four doses) for 12 weeks then placebo orally once per day for 4 weeks; cohort 2: placebo orally once per day for 4 weeks then tolebrutinib orally once per day (one of four doses) for 12 weeks; tolebrutinib doses for both cohorts at 5 mg/day, 15 mg/day, 30 mg/day, or 60 mg/day	Dose-dependent reduction in number of new T1-Gd+ lesions at 12 weeks vs placebo period (mean lesions per patient in the placebo group 1.03 [SD 2.50], in the tolebrutinib 5 mg group 1.39 [3.20], in the tolebrutinib 15 mg 0.77 [1.48], in the 30 mg tolebrutinib group 0.76 [3.31], and in the 60 mg tolebrutinib group 0.13 [0.43]; p=0.03); maximal effect observed with tolebrutinib 60 mg dose (adjusted relative reduction in new T1-Gd+ lesions of 85% [95% CI 28–97]) vs placebo	Occurrence of adverse events was similar across all tolebrutinib groups; only one serious adverse event was reported (one patient in the 60 mg group with a multiple sclerosis relapse); no safety-related discontinuations or treatment-related deaths occurred; the most common adverse event during tolebrutinib treatment was headache (in one [3%] of 13 in the 5 mg group, three [9%] of 15 in the 15 mg group, one [3%] of 33 in the 30 mg group, and four [13%] of 32 in the 60 mg group); three patients had elevated ALT, none required discontinuation of tolebrutinib; other adverse events occurring in at least 5% of patients in any treatment group included upper respiratory tract infection, nasopharyngitis, back pain, peripheral oedema, accidental overdose, gastroenteritis, and muscle spasticity

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Study design	Study population (with summary of key characteristics), total number, and duration	Study groups	Key results: efficacy	Key results: safety
(Continued from previous page)				
Tofebutrinib, NCT03996291²⁸ Long-term (2-year) safety extension of phase 2b trial (NCT03889639)	n=125 patients with relapsing remitting multiple sclerosis who completed phase 2b trial; >92 weeks duration; mean age 37.8 years (SD 9.5); mean disease duration 8.0 years (SD 7.5); median EDSS 2.5 (IQR 1.5–3.5); mean number of relapses in previous year 1.2 (SD 0.6); and 35% of patients with T1–Gd+ lesions on baseline MRI	Part A: participants continued core study tobuterinib dose (5 mg, 15 mg, 30 mg, or 60 mg orally once per day); part B: once 60 mg was identified as the optimal dose, all participants transitioned to open-label tobuterinib 60 mg orally once per day	114 (91%) participants remaining enrolled at 96 weeks; ARR on tobuterinib 60 mg/day for ≥8 weeks was 0.17 (95% CI 0.12–0.25); 81% of participants remained relapse free; median EDSS scores were stable; low but steady rate of new lesion formation in those treated with tobuterinib 60 mg per day (33% with no new or enlarging T2 lesions since long-term safety extension study week 0, and 50% with ≥2 new or enlarging T2 lesions since LTS week 0); in a subgroup of participants (n=29): minimal change in the formation or disappearance of paramagnetic rim lesions: the majority (83%) did not show any change; three (20%) participants showed new PRLs, whereas two (13%) had one fewer PRL over 96 weeks; total volume of SELs in all participants decreased numerically over 96 weeks (0.53 cm³ [95% CI 0.17–1.21] at week 48 and 0.31 cm³ [0.08–0.91] at week 96); additional follow-up reported up to 3.5 years found that participants maintained low ARRs and stable EDSS scores	No new safety signals were identified; the most common TEAEs were COVID-19 (n=26 [20.8%]), headache (n=17 [13.6%]), nasopharyngitis (n=14 [11.2%]), upper respiratory tract infection (n=14 [11.2%]), cystitis/bacterial (n=9 [7.2%]), pharyngitis (n=7 [5.6%]), and arthralgia (n=7 [5.6%]); liver enzyme abnormalities were reported in five (4%) participants; participants recovered without treatment interruption, except one case of increased ALT and AST, who recovered after treatment discontinuation; serious adverse events occurred in 11 (8.8%) of all study participants, with the highest number in the 5 mg and 60 mg group (n=5 [16.1%]); no apparent dose effects were observed for the incidences of TEAEs or serious adverse events; additional follow-up reported up to 3.5 years did not reveal any new safety signals
Tofebutrinib, NCT04410978 (GEMINI1),³⁰ NCT04410991 (GEMINI2)³⁰ Double-blind, double-dummy, event-driven trials	RMS n=1873; median follow-up 139 weeks; mean age in GEMINI 1 with tobuterinib was 36.8 years (SD 9.0), with teriflunomide 36.6 years (9.4) and in GEMINI 2 with tobuterinib 36.6 years (9.3) and teriflunomide 36.1 (9.3); mean time since diagnosis in GEMINI 1 with tobuterinib 4.8 years (SD 6.2), with teriflunomide 4.6 years (SD 6.0) and in GEMINI 2 with tobuterinib 3.8 (5.4) and with teriflunomide 3.8 (5.4); mean EDSS in GEMINI 1 with tobuterinib 2.4 years (1.2) and with teriflunomide 2.4 years (1.2) and in GEMINI 2 with tobuterinib 2.4 years (SD 1.2) and with teriflunomide 2.3 years (1.2); mean number of relapses in previous year in GEMINI 1 with tobuterinib was 1.2 (SD 0.6) and with teriflunomide 1.2 (0.6) and in GEMINI 2 with tobuterinib 1.1 (0.5) and with teriflunomide 1.2 (0.6); 34.7% patients had T1–Gd+ lesions on baseline MRI in GEMINI 1 with tobuterinib and 38.4% with teriflunomide, and in GEMINI 2 32.4% with tobuterinib and 32.6% with teriflunomide	Tofebutrinib 60 mg orally once per day and placebo orally once per day (identical in appearance to teriflunomide); and tobuterinib 14 mg orally once per day and placebo (identical in appearance to tobuterinib)	No differences in ARR between tobuterinib and teriflunomide groups in GEMINI 1 or 2 (0.13 tobuterinib and 0.12 teriflunomide in GEMINI 1 (rate ratio 1.06 [95% CI 0.81–1.39]; p=0.67) and 0.11 in both groups in GEMINI 2 (RR 1.00 [0.75 to 1.32]; p=0.98); pooled 6-month confirmed disability worsening: 8.3% with tobuterinib vs 11.3% with teriflunomide (HR 0.71 [95% CI 0.53–0.95]); pooled 3-month confirmed disability worsening: 11.7% with tobuterinib vs 15.3% with teriflunomide (HR 0.73 [0.57–0.94]); pooled 6-month confirmed disability improvement: 12.6% with tobuterinib vs 10.4% with teriflunomide (HR 1.22 [0.94–1.60]); adjusted mean number of new T1–Gd+ lesions per scan were higher in tobuterinib vs teriflunomide: 0.53 vs 0.29 in GEMINI 1 (ARR 1.86 [95% CI 1.36–2.55]) and 0.46 vs 0.22 in GEMINI 2 (ARR 2.12 [1.50–2.99]); new or enlarging T2 lesions per year were similar across treatment groups: 5.61 with tobuterinib vs 5.18 with teriflunomide in GEMINI 1 (ARR 1.08 [0.88–1.34]) and 5.09 vs 4.37 in GEMINI 2 (ARR 1.17 [0.91–1.50]); least-squares mean percentage change in brain volume was –0.69% with tobuterinib vs –0.88% with teriflunomide in GEMINI 1 (difference 0.20 percentage points [95% CI 0.09 to 0.30]) and –0.70% vs –0.74% in GEMINI 2 (difference 0.04 percentage points [–0.07 to 0.45])	The total number of adverse events observed was similar between the tobuterinib and teriflunomide groups (84.9% vs 86.3%); serious adverse events were seen more commonly with tobuterinib vs teriflunomide (9.8% vs 8.2%); the most common adverse events observed in more than 10% of participants treated with tobuterinib were COVID-19 infection, nasopharyngitis, and headache; the number of liver enzyme elevations of more than three times the upper limit of normal was similar between tobuterinib and teriflunomide treatment groups (5.6% vs 6.3%), but elevations of more than 20 times the upper limit of normal were seen more commonly with tobuterinib vs teriflunomide (0.5% vs 0.1%); n=4 (0.4%) people with liver enzyme elevations of more than three times the upper limit of normal with concurrent bilirubin of more than two times the upper limit of normal in the tobuterinib group; all cases of 20 times the upper limit of normal occurred within 90 days of treatment initiation and resolved without sequelae with treatment discontinuation; one death occurred in the tobuterinib group (homicide) and two deaths in the teriflunomide group (suicide and brain oedema from traumatic subarachnoid haemorrhage); severe infections were similar between groups (2.3% with tobuterinib vs 1.9% with teriflunomide); no substantial differences between groups in moderate-to-severe haemorrhagic events or any other adverse events of special interest

(Table 1 continues on next page)

Study design	Study population (with summary of key characteristics), total number, and duration	Study groups	Key results: efficacy	Key results: safety
(Continued from previous page)				
Tofebutinib, NCT04411641, HERCULES ²¹	Phase 3, randomised, multicentre, double-blind, placebo-controlled, event-driven trial n=1131 participants with secondary progressive multiple sclerosis with no superimposed relapses in previous 2 years; median follow-up 133 weeks; mean age 48.9 (SD 8.0) for both groups; mean relapsing remitting multiple sclerosis disease duration with tofebutinib 17.1 years (8.3) vs placebo 17.6 years (8.4); mean time since SPMS secondary progressive multiple sclerosis diagnosis with tofebutinib was 7.9 years (7.3) vs placebo 8.4 years (7.8); mean EDSS with tofebutinib was 5.5 (SD 1.0) vs placebo 5.6 (0.9); mean time since last relapse with tofebutinib was 7.4 (5.3) vs placebo 7.6 (5.5); 12.5% patients had T1-Gd+ lesions on baseline MRI with tofebutinib and 13.1% with placebo	Tofebutinib 60 mg orally once per day and matched placebo orally once per day 6-month CDP was seen less frequently with tofebutinib vs placebo: 22.6% vs 30.7% (HR 0.69 [95% CI 0.55–0.88]; p=0.003); 3-month CDP was seen less frequently with tofebutinib vs placebo: 27.6% vs 34.2% (HR 0.76 [0.61–0.94]; p=0.01); 9 hole peg test worsening sustained ≥3 months was similar between tofebutinib and placebo: 19.0% vs 19.6% (HR 0.97 [0.74–1.29]; p=0.84); timed 25-foot walk worsening sustained ≥3 months was seen less frequently with tofebutinib vs placebo: 41.1% vs 49.6% (HR 0.77 [0.64–0.92]); 6-month CDI was seen more frequently with tofebutinib vs placebo: 8.6% vs 4.5% (HR 1.88 [1.10–3.21]; mean annualised rate of new or enlarging T2 lesions: 1.84 with tofebutinib vs 2.95 with placebo (RR 0.62 [95% CI 0.43–0.90]; p=0.01); least-squares mean percentage change in brain volume: –0.69% with tofebutinib vs –0.78% with placebo (least-squares mean difference 0.08 [95% CI –0.03 to 0.20])	The total number of adverse events were higher in the tofebutinib group vs the placebo group (81.5% vs 78.1%); adverse events were reported in at least 10% of patients receiving tofebutinib, including COVID-19 infection and urinary tract infections; respiratory infections were more common with tofebutinib vs placebo, including COVID-19 (25.5% vs 22.7%), nasopharyngitis (9.3% vs 6.9%), and influenza (5.6% vs 3.5%); serious adverse events occurred more commonly in tofebutinib vs placebo (15.0% vs 10.4%); the most common serious adverse events with tofebutinib were COVID-19 pneumonia (1.1%), multiple sclerosis relapse (1.1%), COVID-19 (0.9%), and pneumonia (0.7%); elevations in liver enzymes were reported more frequently in tofebutinib vs placebo (4% vs 1.6%); elevations more than 20 times the upper limit of normal were seen only in the tofebutinib group (0.5%); three of these cases (0.4%) had concurrent elevation of bilirubin of more than two times the upper limit of normal; one death occurred in the tofebutinib group related to postoperative complications of a liver transplant; underlying liver failure was deemed to be related to tofebutinib; after the implementation of weekly liver enzyme monitoring, subsequent cases of severe liver enzyme elevation resolved without sequelae	
Fenebutinib, NCT05119569, FENOPTA ²²	Phase 2, randomised, multicentre, double-blind, placebo-controlled trial n=109 = relapsing remitting multiple sclerosis; 12 weeks double-blind placebo; optional OLE up to 192 weeks; mean age was 39.0 (SD 8.2); median time since diagnosis was 2.0 years (IQR 0.5–8.2); median EDSS 2.5 (IQR 2.0–3.0); median time since last relapse was 4.3 months (IQR 2.6–6.5); 19% of patients had T1-Gd+ lesions on baseline MRI	Fenebutinib 200 mg orally twice per day; placebo orally twice per day; and optional OLE fenebutinib 200 mg orally twice per day Combined new T1-Gd+ lesions at weeks 4, 8, and 12 were lower with fenebutinib vs placebo amounting to a 69% relative reduction (95% CI 34–85; p=0.0022); at weeks 4, 8, and 12 there was a relative reduction of new T1-Gd+ lesions with fenebutinib vs placebo: 22% relative reduction (95% CI –92 to 68) at week 4, 92% (65 to 98) at week 8, and 90% (52 to 98) at week 12; combined new or enlarging T2 lesions at weeks 4, 8, and 12 (fenebutinib vs placebo): 74% relative reduction (95% CI 45–87; p=0.0004); at OLE week 48: mean number of new T1-Gd+ lesions was 0.03 lesions per scan (n=89); 0.05 lesions in the fenebutinib–fenebutinib cohort and 0 in the placebo–fenebutinib cohort; at OLE week 48, the annualised rate of new or enlarging T2 lesions was 0.17 (95% CI 0.06–0.46) in the fenebutinib–fenebutinib group and 0.22 (0.062–0.79) in the placebo–fenebutinib group; unadjusted ARR during double-blind placebo: 0 with fenebutinib vs 0.18 with placebo; at OLE week 48 ARR was 0.04 overall, 0.2 with fenebutinib–fenebutinib vs 0.1 with placebo–fenebutinib	No serious adverse events or deaths were reported during the double-blind placebo phase; adverse events occurring in at least 5% of participants were urinary tract infection (four [5.5%] with fenebutinib vs two [5.6%] placebo), and abnormal liver enzymes (4 [5.5%] vs 0); n=6 (8.2%) of patients treated with fenebutinib withdrew because of an adverse event (three because of ALT or AST elevations of more than eight times the upper limit of normal; one because of upper abdominal pain, nausea, and headache; one because of upper abdominal pain, and one because of hypersensitivity); liver enzyme elevations were reversible and asymptomatic; during the OLE phase, one patient had two serious adverse events, nephrolithiasis and urinary tract infections; the most common adverse events in more than 5% of patients in OLE were urinary tract infections (8%), COVID-19 (7%), and pharyngitis (5%); an asymptomatic ALT elevation newly occurred in one patient (1%) at OLE week 4 (after 16 weeks of fenebutinib treatment) and resolved with treatment discontinuation	

ARR=annualised relapse rate. EDSS=expanded disability status scale. CSF=cerebrospinal fluid. BTKI=Bruton's tyrosine kinase inhibitor. Nfl=neurofilament light chain. OLE=open-label extension. TEAE=treatment-emergent adverse event. T1-Gd+=T1 gadolinium-enhancing.

Table 1: Summary of phase 2 and 3 clinical trials of BTKIs in multiple sclerosis with published results

ARR=annualised relapse rate. EDSS=expanded disability status scale. CSF=cerebrospinal fluid. BTKI=Bruton's tyrosine kinase inhibitor. NFL=neurofilament light chain. OLE=open-label extension. T1-Gd+=T1 gadolinium-enhancing.

Table 1: Summary of phase 2 and 3 clinical trials of BTKIs in multiple sclerosis with published results

trials that assessed evobrutinib versus teriflunomide in people with relapsing multiple sclerosis. At a median of 109 weeks, there was no difference observed in annualised relapse rates between treatment groups in evolution RMS 1 or 2,²⁷ which were comparably low (table 1). Secondary clinical endpoints, including 12-week and

24-week confirmed disability progression and 24-week confirmed disability improvement, were not different between treatment groups (table 1).²⁷ MRI endpoints, including the total number of T1-Gd+ lesions and adjusted mean number of T1-Gd+ lesions per scan, were numerically higher for evobrutinib versus teriflunomide

	Study design	Study population, total number, and duration	Study groups	Key results: efficacy	Key results: safety
Tolebrutinib, PERSEUS (NCT04458051)	Phase 3, randomised, multicentre, double-blind, efficacy and safety study	Primary progressive multiple sclerosis (n=767 actual enrolment); treatment duration was variable (event-driven study)	Tolebrutinib 60 mg orally once per day and placebo orally once per day	Information based on conference abstract: tolebrutinib did not delay time to onset of 6-month composite confirmed disability progression vs placebo (HR 1.01 [0.81–1.26]; p=0.94)	Information based on conference abstract: safety was consistent with previous phase 3 trials of tolebrutinib
Tolebrutinib, (NCT06372145)	Phase 3, multicentre, single-group, open-label extension study	Relapsing multiple sclerosis, non-relapsing secondary progressive multiple sclerosis, or primary progressive multiple sclerosis previously enrolled in the following studies: NCT03996291, NCT04410978 (GEMINI 1), NCT04410991 (GEMINI 2), NCT04411641 (HERCULES), or NCT04458051 (PERSEUS); n=2500 (estimated); 3 years duration	If previously in NCT03996291, continued open-label tolebrutinib 60 mg orally once per day; if previously in NCT04410978 or NCT04410991 on teriflunomide, will undergo accelerated elimination or 3-month washout period, then started on open-label tolebrutinib 60 mg orally once per day; if previously in NCT04410978 or NCT04410991 on teriflunomide and were benefiting and if recommended, can continue on teriflunomide 14 mg orally once per day outside of the current study; if previously in NCT04410978 or NCT04410991 on tolebrutinib, will receive open-label tolebrutinib 60 mg orally once per day; if previously in NCT04411641 or NCT04458051, will receive open-label tolebrutinib 60 mg orally once per day; if tolebrutinib open label was started in any parent study, will continue 60 mg orally once per day	Expected in 2029	Expected in 2029
Fenebrutinib, FENhance (NCT04586010), FENhance 2 (NCT04586023)	Phase 3 randomised, multicentre, double-blind, double-dummy, parallel-group study	Relapsing multiple sclerosis (n=751 actual enrolment)	Fenebrutinib 200 mg orally twice per day; teriflunomide 14 mg orally once per day; possible OLE (after DBT results)	Topline results announced for FENhance; FENhance 2 results expected in 2027	Expected in 2027
Fenebrutinib, FENTrepid (NCT04544449)	Phase 3 randomised, multicentre, double-blind, double-dummy, parallel-group study	Primary progressive multiple sclerosis (n=985 actual enrolment)	Fenebrutinib 200 mg orally twice per day; ocrelizumab 600 mg intravenously every 6 months; possible OLE (after DBT results)	Information based on the conference abstract: fenebrutinib was non-inferior to ocrelizumab at delaying the onset of composite confirmed disability progression over a period of at least 120 weeks (HR 0.88 [95% CI 0.75–1.03], synthesis test statistic –3.02); a numerical benefit for fenebrutinib vs ocrelizumab was seen as early as week 24, and lasted throughout the observation period	Information based on the conference abstract: overall safety profile was similar to ocrelizumab, imbalance in liver enzyme elevations and fatalities observed

(Table 2 continues on next page)

Study design		Study population, total number, and duration	Study groups	Key results: efficacy	Key results: safety
(Continued from previous page)					
Remibrutinib, REMODEL-1 (NCT05147220), REMODEL-2 (NCT05156281)	Phase 3, randomised, multicentre, double-blind, double-dummy, active comparator-controlled, fixed-dose, parallel-group study, followed by OLE	Relapsing multiple sclerosis (n=800 estimated); core study up to 30 months, OLE study up to 5 years	Remibrutinib 100 mg orally twice per day; teriflunomide 14 mg orally once per day; remibrutinib orally (single-group OLE), previously on remibrutinib or teriflunomide	Expected in 2030	Expected in 2030
Remibrutinib, (NCT06846281)	Phase 3b, randomised, open-label, parallel-group, non-inferiority study	Relapsing multiple sclerosis ≥40 years; treated with ocrelizumab according to routine clinical practice and at standard dose, neurological stability in previous 30 days, suitable to be switched to remibrutinib based on physician judgement or patient preference; n=360 (estimated); 24 months duration (core) and 24 months duration (OLE)	Remibrutinib 100 mg orally twice per day (core study); ocrelizumab 600 mg intravenously for 6 months or 920 mg subcutaneously every 6 months (core study); remibrutinib orally (OLE study, on remibrutinib in core study); and remibrutinib orally (OLE study, on ocrelizumab in core study)	Expected in 2031	Expected in 2031
Remibrutinib, (NCT07225504)	Phase 3, randomised, double-blind, placebo-controlled, multicenter, parallel-group, event-driven study	Secondary progressive multiple sclerosis (n=1275 estimated); remibrutinib vs placebo event driven); followed by extension (open-label remibrutinib)	Remibrutinib (dose not indicated); placebo	Expected in 2034	Expected in 2034
Orelabrutinib, (NCT04711148)	Phase 2, randomised, multicentre, double-blind, placebo-controlled study, followed by OLE	Relapsing multiple sclerosis (n=160 estimated); core study was 24 weeks, OLE study was 120 weeks	Placebo; orelabrutinib 50 mg once per day; orelabrutinib 80 mg once per day; and orelabrutinib 50 mg twice per day	Information based on the conference abstract: all three treatment groups showed significant reductions in the cumulative number of new T1-Gd+ lesions and new or enlarging T2 lesions vs placebo at week 12 (p<0.05); the group on 80 mg per day demonstrated the highest reductions of 90.4% at week 12 vs placebo; the effect of orelabrutinib on new lesions was seen as early as week 4, sustained through 24 weeks	Information based on the conference abstract: proportion of participants with TEAEs was higher with orelabrutinib (8–60%) vs placebo (30%); serious adverse events were seen in 4.9% of participants in highest-dose orelabrutinib group, none in other groups; liver-related TEAEs were seen in 5–13.5% of participants in orelabrutinib groups, and 7.5% in the placebo group
Orelabrutinib, (NCT07067463)	Phase 3, randomised, double-blind, multicentre, placebo-controlled study	Primary progressive multiple sclerosis (estimated n=705)	Orelabrutinib (dose not indicated); placebo	Expected in 2030	Expected in 2030

(Table 2 continues on next page)

	Study design	Study population, total number, and duration	Study groups	Key results: efficacy	Key results: safety
(Continued from previous page)					
BIIB091, (NCT05798520)	Phase 2, two-part, randomised, multicentre, blinded, active-controlled study	Relapsing multiple sclerosis (n=275 estimated); 48 weeks duration (part 1), and 48 weeks duration (part 2)	BIIB091 (high dose) orally twice per day and diroximel fumarate-matching placebo orally twice per day (part 1); BIIB091 (low dose) orally twice per day and diroximel fumarate-matching placebo orally twice per day (part 1); diroximel fumarate (standard dose) orally twice per day and BIIB091-matching placebo orally twice per day (part 1); BIIB091 orally twice per day + diroximel fumarate (standard dose) orally twice per day (part 2); BIIB091 orally twice per day + diroximel fumarate (low dose) orally twice per day (part 2); and diroximel fumarate (standard dose) orally twice per day and BIIB091-matching placebo orally twice per day (dose of BIIB091 in groups 4 and 5 will be determined by results from part 1)	Expected in 2026	Expected in 2026
BTKI=Bruton's tyrosine kinase inhibitor. OLE=open-label extension. T1-Gd+=T1 gadolinium-enhancing.					
Table 2: Summary of phase 2 and 3 clinical trials of BTKIs in multiple sclerosis that are without published results or ongoing					

in both evolution RMS 1 and 2²⁷ (table 1). The number of new or enlarging T2 lesions was similar between treatment groups in both studies and did not increase after week 24. Tertiary and prespecified endpoints of interest, including progression independent of relapsing activity, whole-brain volume, and paramagnetic rim lesion count were not different between treatment groups.²⁷ However, the slowly expanding lesion volume was numerically lower with evobrutinib versus teriflunomide at week 96, whereas there was no effect on confirmed disability progression (table 1).²⁷ Serious adverse events were more common with evobrutinib than with teriflunomide (7·5% vs 5·6%), and liver enzyme elevations more than five times the ULN were more common with evobrutinib, particularly in the first 12 weeks (5% with evobrutinib vs <1% with teriflunomide). Three participants in the evobrutinib group met the biochemical definition of Hy's Law; all three were asymptomatic, and their liver enzymes normalised after treatment discontinuation.³⁴ These participants led to a regulatory hold of evolution RMS 1 and 2,²⁷ and frequent liver enzyme monitoring was subsequently implemented in the trial protocol (table 1).²⁷

Results of three phase 3 trials of tolebrutinib in relapsing and non-relapsing secondary progressive multiple sclerosis have also been reported, with key differences observed when compared to the evobrutinib phase 3 trials.^{27,30,31} GEMINI 1 and 2³⁰ were two phase 3 trials that evaluated the effect of tolebrutinib at 60 mg per day versus teriflunomide at 14 mg per day in people with relapsing multiple sclerosis. At a median follow-up

of 139 weeks, the primary outcome was not different between treatment groups, with similarly low annualised relapse rates (table 1).³⁰ Secondary endpoints, including 3-month and 6-month confirmed disability worsening, were less frequently in participants treated with tolebrutinib (table 1). T1-Gd+ lesions were more frequent with tolebrutinib versus teriflunomide, but new or enlarging T2-lesions were similar across treatment groups (table 1).³⁰ The rate of whole-brain brain atrophy from month 6 to the end of the trial was lower with tolebrutinib versus teriflunomide in GEMINI 1 but not GEMINI 2.³⁰ Notably, the observed rates of brain atrophy across treatment groups were within the range of what is observed in healthy controls (table 1).³⁰ In GEMINI 1 and 2,³⁰ the proportions of serious adverse events were similar with tolebrutinib versus teriflunomide (9·8% vs 8·2%). The number of liver enzyme elevations of more than three times the ULN was similar between tolebrutinib versus teriflunomide treatment groups (5·6% vs 6·3%), but elevations of more than 20 times the ULN were more common with tolebrutinib versus teriflunomide (0·5% vs 0·1%). There were four people (0·4%) that met biochemical Hy's law; all occurred within 90 days of treatment initiation and resolved without sequelae with treatment discontinuation (table 1).³⁰

HERCULES³¹ was a phase 3 trial that evaluated the effect of tolebrutinib at 60 mg per day versus placebo in people with non-relapsing secondary progressive multiple sclerosis. At a median follow-up of 133 weeks, participants in the tolebrutinib group showed a reduced

risk of 6-month confirmed disability progression (table 1). Similarly, there was a reduced risk of 3-month confirmed disability progression in the tolebrutinib group (table 1). The 9 hole peg test (9HPT) was not different across treatment groups, thus subsequent secondary endpoints were considered to be of nominal importance; however, an apparently lower proportion of participants in the tolebrutinib group showed worsening of 25 foot timed walk, and an apparently higher proportion demonstrated 6-month confirmed disability improvement (table 1). The mean annualised rate of new or enlarging T2 lesions was lower with tolebrutinib versus placebo, whereas the least-squares mean percentage change in brain volume at the end of the trial compared with month 6 was not different across treatment groups (table 1).³¹ In HERCULES,³¹ the number of participants with serious adverse events was higher in the tolebrutinib group versus the placebo group (15% vs 10·4%; table 1). Elevations in liver enzymes were more frequent in tolebrutinib versus placebo (4% vs 1·6%), and elevations of more than 20 times the ULN were seen only in the tolebrutinib group (0·5%). Three (0·4%) of these participants met biochemical Hy's law. There was one death in the tolebrutinib group related to postoperative complications of a liver transplant; underlying liver failure was deemed related to tolebrutinib. Of note, this participant developed liver failure before the frequent liver monitoring protocol was implemented. After weekly liver enzyme monitoring was implemented, subsequent elevations in liver enzymes resolved without sequelae (table 1).³¹

The observation of more T1-Gd+ lesions with both tolebrutinib and evobrutinib relative to teriflunomide is surprising, as there were no increases in clinical relapses or new or enlarging T2 lesions. One can speculate that this observation might be related to increased or more durable changes in vascular permeability unrelated to the development of new peripherally mediated disease activity. Possible explanations for the absence of an effect of tolebrutinib on whole-brain atrophy versus placebo in HERCULES, despite demonstrating an effect on confirmed disability progression, include that whole-brain atrophy is a non-specific and delayed measure reflecting tissue degeneration due to both multiple sclerosis-related pathophysiological processes and other causes, including normal ageing and comorbidities. As a result, it is possible that a clinical benefit related to attenuating the contribution of multiple sclerosis pathophysiological processes might not correspond to a decrease in the rate of brain atrophy, particularly in the HERCULES clinical trial population, which was primarily composed of people with non-relapsing SPMS, a population not well represented in previous multiple sclerosis clinical trials. Moreover, nearly half of participants in HERCULES were previously on multiple sclerosis treatment, many with long-lasting benefits, including anti-CD20

monoclonal antibodies and immune reconstitution therapies, which might have already diminished the rate of whole-brain atrophy before they started tolebrutinib.³¹

Recent post-hoc prespecified analyses presented as conference abstracts have provided further insights into the effects of tolebrutinib. In GEMINI 1 and 2,³¹ tolebrutinib-treated participants had a reduced risk of progression independent of relapsing activity compared with those treated with teriflunomide (9·9% vs 7·6%, HR 0·73 [0·52–1·02]).³⁷ Furthermore, a post-hoc analysis of baseline paramagnetic rim lesions in a subgroup of participants from HERCULES and GEMINI 1 and 2³⁰ (about 34–39%) found that baseline paramagnetic rim lesion count predicted confirmed disability progression over the study period in both the placebo and teriflunomide treatment groups.⁵⁰ Specifically, in those with more than four paramagnetic rim lesions, there was a 54% risk reduction of confirmed disability progression in tolebrutinib versus placebo in HERCULES and a 49% risk reduction of confirmed disability worsening in tolebrutinib versus teriflunomide in GEMINI 1 and 2.³⁰ Although requiring confirmation in future clinical trials, these findings highlight the prognostic utility of paramagnetic rim lesions in predicting confirmed disability progression over 2–3 years in untreated people with multiple sclerosis or those on teriflunomide and raise the possibility that paramagnetic rim lesions might be a predictive biomarker of treatment response to tolebrutinib.³⁸

Clinical trials that are without published results or ongoing

Recently, results of two phase 3 trials of fenebrutinib and a phase 3 trial of tolebrutinib were reported in abstract form or by press release: FENhance 2³⁹ (fenebrutinib vs teriflunomide in relapsing multiple sclerosis), FENTrepid⁴⁰ (fenebrutinib vs ocrelizumab in primary progressive multiple sclerosis), and PERSEUS⁴¹ (tolebrutinib vs placebo in primary progressive multiple sclerosis; table 2). Encouraging topline results for the FENhance 2 trial have been announced.³⁹ FENTrepid demonstrated that fenebrutinib is non-inferior in reducing composite confirmed disability progression versus ocrelizumab (table 2).⁴⁰ PERSEUS did not meet its primary endpoint, with no difference reported in the primary outcome of 6-month composite confirmed disability progression between tolebrutinib versus placebo (table 2).⁴¹ Published and additional results from these trials, as well as the results of FENhance (NCT04586010), are pending (table 2). There are ongoing phase 2 and 3 clinical trials evaluating the safety and efficacy of remibrutinib (NCT06846281 and NCT07225504), orelabrutinib (NCT07067463), and BIIB091 (NCT05798520) that do not yet have any reported results (table 2).

Differences among BTKIs in development for multiple sclerosis

Differences in pharmacology

BTKIs under development in multiple sclerosis harbour differences in a range of pharmacological properties, including binding mechanism, reversibility, selectivity, potency, pharmacokinetic and pharmacodynamic properties, and CNS penetrance (appendix pp 1–2).

See Online for appendix

CNS penetrance is of particular interest for BTKIs in multiple sclerosis. Rapid CNS penetrance within hours of oral ingestion has been observed with several BTKIs on the basis of CSF concentrations, which might not necessarily reflect CNS tissue concentrations. However, there appear to be differences in the ability to achieve biologically relevant concentrations in the CNS.^{26,32,42}

In a subset of nine people with multiple sclerosis on evobrutinib 75 mg twice per day in the phase 2 trial,²⁶ CSF concentrations were 3.2 ng/mL (range 1.3–5.2) 2 h after administration, below the reported half-maximal inhibitory concentration (IC_{50}) for BTK occupancy (3.8 ng/mL),⁴³ suggesting that the CNS penetrance of evobrutinib is probably below the levels required to modulate responses of relevant cells within the CNS at the doses studied. In healthy volunteers, after a single dose of 120 mg, tolebrutinib was present in the CSF at 1.87 ng/mL (4.1 nM), which is higher than the IC_{50} for BTK occupancy (0.3 ng/mL [0.7 nM]) and the IC_{50} to inhibit microglia (0.32 ng/mL [0.7 nM]).^{42,44} Similarly, fenebrutinib has shown CNS penetrance at bioactive concentrations; in a CSF substudy of 11 people with multiple sclerosis in the phase 2 trial, fenebrutinib was detectable in all participants at week 12, at 43.1 ng/mL (64.8 nM), exceeding the IC_{50} for BTK phosphorylation (2.3 nM [1.53 ng/mL])⁴⁵ and for downstream activation markers (CD63 in a myeloid cell line 3.00 nM [2.0 ng/mL] and CD69 in a B-cell line 19.8 nM [13.16 ng/mL]).^{32,46–48} Remibrutinib has also demonstrated evidence of CNS penetrance at biologically relevant concentrations; in 12 healthy volunteers, within 2 h after ingestion, remibrutinib was detectable at 1.54 nM (range 1.03–1.87) or 0.78 ng/mL (range 0.52–0.95), which is above the IC_{50} for microglia inhibition (1.1 nM [0.56 ng/mL]).^{22,49}

Selectivity is another property that might result in differences in safety and efficacy, as illustrated by the observed adverse event profile of ibrutinib, a first-generation BTKI with known off-target effects that can cause serious bleeding and cardiac events.^{50–53} Of BTKIs under investigation in multiple sclerosis, remibrutinib has shown the least off-target kinase activity (on the basis of platforms designed to evaluate each BTKIs' inhibition of BTK and other off-target human kinases), followed by fenebrutinib, evobrutinib, and tolebrutinib.^{42,49,50,53,54} The binding mechanism and reversibility are also properties that differ amongst BTKIs being evaluated in multiple sclerosis, with only fenebrutinib and BIIB091 showing non-covalent

binding and reversibility.^{42,54,55–57} Although binding mechanism does not necessarily affect BTKI reversibility given the rapid regeneration of BTK with discontinuation, covalent binding has been found to result in development of tolerance to BTKIs in the context of malignancy.^{58,59}

Differences in clinical trial results

The differences observed in T1-Gd+ lesion suppression in the phase 2 open-label extension of fenebrutinib³² in comparison with phase 2 trials of evobrutinib²⁵ and tolebrutinib²⁸ are of interest, as are the differences observed in effect on annualised relapse rate and confirmed disability progression in the phase 3 evobrutinib,²⁷ tolebrutinib,³⁰ and fenebrutinib³⁹ trials. Although speculative, possible reasons for the apparent greater effect of fenebrutinib on relapsing disease biology might include key differences in dosing, pharmacokinetic profiles, and potencies, which might result in persistently high serum concentrations of free fenebrutinib. This effect might then translate to differences in the magnitude of effect on B cells in the periphery, resulting in differences in the magnitude of suppression of T1-Gd+ lesions. Furthermore, the unique binding mechanism of fenebrutinib (non-covalent and reversible) in comparison with evobrutinib and tolebrutinib (covalent and irreversible) could result in a more persistent effect on B cells because of the absence of development of tolerance to BTK inhibition over time that has been reported in the oncology literature.^{58,59}

The discrepancy between results of evolution RMS 1 and 2²⁷ and GEMINI 1 and 2,³⁰ particularly considering the positive results of the HERCULES trial,³¹ are of interest. A possible explanation is what is known about the differences in CNS penetrance of evobrutinib and tolebrutinib, specifically the ability of tolebrutinib to achieve biologically relevant concentrations in the CNS.

Although difficult to speculate as to why PERSEUS⁴¹ did not meet its primary endpoint with limited available results reported in a conference abstract, it is possible that PERSEUS recruited a population with milder primary progressive multiple sclerosis because of the competing recruitment with FENTREPID⁴⁰ in which both groups received active treatment. Another possibility is that the primary endpoint of composite confirmed disability progression was not sensitive enough to detect a difference between treatment groups over the study duration. Moreover, the overall pattern of findings appears broadly consistent across PERSEUS and HERCULES, as tolebrutinib was associated with effects on disability accumulation and brain atrophy that were directionally aligned across trials, which raises the possibility that the apparent differences at the level of individual trials might be attributable to random variation around relatively small effect sizes. Forthcoming publications detailing PERSEUS and FENTREPID⁴⁰ results will be important to better

understand this discrepancy. Forthcoming results of other ongoing phase 2 and 3 trials (table 2) and real-world evidence will better inform relevant differences of BTKIs with respect to their effect on both relapsing and progressive disease processes and their clinical implications.

Reflections on the use of BTKIs in multiple sclerosis clinical practice

If replicated in trials expected to be reported over the next several years, results from the tolebrutinib and fenebrutinib phase 3 clinical trials will mark a new milestone in the multiple sclerosis field. Before the HERCULES³¹ and GEMINI 1 and 2³⁰ clinical trials of tolebrutinib, no previous therapy had demonstrated an effect on disability progression that is largely independent of relapsing disease activity. Considering what is known about the abilities of tolebrutinib and fenebrutinib to penetrate the CNS at biologically relevant concentrations, these observations might possibly be the harbingers of an era of disease-modifying treatments that can meaningfully modulate CNS-compartmentalised disease processes relevant to progressive disease biology. Given the negative results of PERSEUS⁴¹ (conference abstract), it will be important to understand why tolebrutinib did not show an effect in primary progressive multiple sclerosis and future analyses will hopefully facilitate an understanding of these discrepant results. Nonetheless, such DMTs could have clinical value in multiple sclerosis and strengthen the argument that multiple sclerosis should be considered a single disease entity with variable clinical manifestations and that treatments should be geared to multiple sclerosis biology rather than currently defined clinical subtypes.⁷ The optimal patient population for specific BTKIs might differ and will be informed by upcoming clinical trial results with respect to magnitude of effect on clinical, radiological, and fluid biomarker evidence of relapsing and progressive disease biology. Novel treatment strategies using BTKIs in combination with other DMTs, or in rapid sequence after anti-CD20 monoclonal antibody therapy, are intriguing possibilities that have the potential to address unmet clinical needs. Early results from a small, single-centre phase 2a trial⁶⁰ that sequenced tolebrutinib after anti-CD20 monoclonal antibody therapy in people with multiple sclerosis identified no safety concerns. At 48 weeks, the primary outcome (disappearance of paramagnetic rim lesions on 7T MRI) was not observed in either tolebrutinib (n=7) or the reference group (participants continuing anti-CD20 monoclonal antibodies; n=5). However, pilot biomarker data suggested a benefit for non-relapsing multiple sclerosis disease activity.^{60,61}

With regards to safety, the off-target effects seen with first-generation BTKIs, including severe bleeding and cardiac arrhythmias, have not been observed to date in

Search strategy and selection criteria

We identified references for this Review by searching PubMed and references of relevant articles in the English language using the search terms “multiple sclerosis”, “BTK-inhibitor”, “Bruton’s tyrosine kinase inhibitor”, “clinical trials”, and “pre-clinical” from Jan 1, 2019 to Feb 7, 2026. We also searched abstract databases from recent meetings, including the European Committee for Treatment and Research in Multiple Sclerosis 2019–25, the American Academy of Neurology 2019–25, and the Americas Committee for Treatment and Research in Multiple Sclerosis 2019–26, European Academy of Neurology 2019–25, International Society of Neuroimmunology 2019–25, and Consortium of Multiple Sclerosis Centers 2019–25. We also used ClinicalTrials.gov to identify relevant clinical trials. The final reference list was compiled based on relevance to the scope of this Review.

multiple sclerosis clinical trials. The underlying mechanisms of the liver injury seen with many BTKIs are not yet clear, but this safety concern seems to be shared amongst many, but not all, BTKIs being evaluated in multiple sclerosis. Rare occurrences of drug-induced liver injury seen with ibrutinib provide some insights into the potential mechanisms of liver injury related to BTKIs: pathological specimens have shown evidence of both hepatocellular and cholestatic changes, and inflammatory infiltrates, together suggesting an idiosyncratic reaction that possibly involves both inflammatory and toxic mediators.^{62,63} Should BTKIs be approved for use in multiple sclerosis, frequent laboratory monitoring will probably be required for many weeks after drug initiation, followed by intermittent monitoring thereafter. A better understanding of the short-term and long-term safety profile of BTKIs will be an important consideration for use in clinical practice. Much needed insights into the effectiveness of risk mitigation strategies for liver safety, and whether this concern is a class effect of BTKIs, will be provided through additional clinical trials, long-term extension studies, and ongoing pharmacovigilance.

Although immune responses to vaccines have not been evaluated in people with multiple sclerosis participating in BTKI clinical trials, studies from BTKI use in haematological malignancy suggest that humoral immune responses to de-novo antigens might be impaired with the use of BTKIs. As such, this is an area requiring further study in multiple sclerosis.^{64,65}

Tolebrutinib received regulatory approval in the United Arab Emirates, but the US Food and Drug Administration did not grant approval and issued a complete response letter, requesting for additional data.^{66,67} Should BTKIs eventually be approved, their pricing and potentially demanding schedules for liver monitoring will be important factors in their access and clinical uptake in

both high-resource and low-resource countries. Finally, the results of ongoing and upcoming clinical trials will clearly inform the place of BTKIs in multiple sclerosis clinical practice and provide additional evidence that will guide a nuanced approach to determining the patient population that will optimally benefit most from these drugs.

Conclusions and future directions

BTKIs are a new class of DMT in multiple sclerosis that hold promise for addressing several unmet needs in multiple sclerosis clinical practice. Recently reported phase 3 clinical trials have provided the field with renewed hope that some BTKIs might be able to modulate progressive multiple sclerosis disease processes in specific multiple sclerosis populations that current DMTs do not effectively treat. Differences in specific pharmacological properties of BTKIs might contribute to safety and efficacy differences that are of high relevance clinically. Research priorities include confirming the safety and efficacy of BTKIs in people with multiple sclerosis with results from upcoming clinical trials and real-world studies, understanding why an effect might have been seen in non-relapsing secondary progressive multiple sclerosis but not primary progressive multiple sclerosis with tolebrutinib, and confirming the mechanisms by which BTKIs beneficially modulate specific disease pathways of clinical relevance in multiple sclerosis. These data and regulatory approval decisions are eagerly anticipated and will guide if and how BTKIs will be used in clinical practice to improve outcomes of people living with multiple sclerosis.

Contributors

JO did the study conceptualisation, literature search, data collection, data analysis, data interpretation, and writing (original draft and review and editing). AB-O, XM, M-PS, and MSW did the data analysis, data interpretation, and writing (review and editing). DSR did the literature search, data collection, data analysis, data interpretation, and writing (review and editing).

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References

- 1 Brunner C, Müller B, Wirth T. Bruton's tyrosine kinase is involved in innate and adaptive immunity. *Histol Histopathol* 2005; **20**: 945–55.
- 2 Vetrie D, Vorechovsky I, Sideras P, et al. The gene involved in X-linked agammaglobulinaemia is a member of the src family of protein-tyrosine kinases. *Nature* 1993; **361**: 226–33.
- 3 Hendriks RW, Yuvaraj S, Kil LP. Targeting Bruton's tyrosine kinase in B cell malignancies. *Nat Rev Cancer* 2014; **14**: 219–32.
- 4 Stanchina MD, Montoya S, Danilov AV, et al. Navigating the changing landscape of BTK-targeted therapies for B cell lymphomas and chronic lymphocytic leukaemia. *Nat Rev Clin Oncol* 2024; **21**: 867–87.
- 5 Isaacs JD. Bruton's tyrosine kinase: a new target for immune mediated inflammatory diseases? *Semin Arthritis Rheum* 2025; **72S**: 152681.
- 6 Reich DS, Lucchinetti CF, Calabresi PA. Multiple sclerosis. *N Engl J Med* 2018; **378**: 169–80.
- 7 Kuhlmann T, Moccia M, Coetzee T, et al, and the International Advisory Committee on Clinical Trials in Multiple Sclerosis. Multiple sclerosis progression: time for a new mechanism-driven framework. *Lancet Neurol* 2023; **22**: 78–88.
- 8 Bar-Or A, Li R. Cellular immunology of relapsing multiple sclerosis: interactions, checks, and balances. *Lancet Neurol* 2021; **20**: 470–83.
- 9 Dybowski S, Torke S, Weber MS, Targeting B. Targeting B cells and microglia in multiple sclerosis with Bruton tyrosine kinase inhibitors: a review. *JAMA Neurol* 2023; **80**: 404–14.
- 10 Geladaris A, Häusler D, Weber MS. Microglia: The missing link to decipher and therapeutically control MS progression? *Int J Mol Sci* 2021; **22**: 3461.
- 11 Kappos L, Wolinsky JS, Giovannoni G, et al. Contribution of relapse-independent progression vs relapse-associated worsening to overall confirmed disability accumulation in typical relapsing multiple sclerosis in a pooled analysis of 2 randomized clinical trials. *JAMA Neurol* 2020; **77**: 1132–40.
- 12 Yong HYF, Yong VW. Mechanism-based criteria to improve therapeutic outcomes in progressive multiple sclerosis. *Nat Rev Neurol* 2022; **18**: 40–55.
- 13 Oh J, Bar-Or A. Emerging therapies to target CNS pathophysiology in multiple sclerosis. *Nat Rev Neurol* 2022; **18**: 466–75.

- 14 Geladakis A, Torke S, Weber MS. Bruton's tyrosine kinase inhibitors in multiple sclerosis: pioneering the path towards treatment of progression? *CNS Drugs* 2022; **36**: 1019–30.
- 15 Gruber RC, Wirak GS, Blazier AS, et al. BTK regulates microglial function and neuroinflammation in human stem cell models and mouse models of multiple sclerosis. *Nat Commun* 2024; **15**: 10116.
- 16 Geladakis A, Torke S, Saberi D, et al. BTK inhibition limits microglia-perpetuated CNS inflammation and promotes myelin repair. *Acta Neuropathol* 2024; **147**: 75.
- 17 Bame E, Tang H, Burns JC, et al. Next-generation Bruton's tyrosine kinase inhibitor B1B091 selectively and potently inhibits B cell and Fc receptor signaling and downstream functions in B cells and myeloid cells. *Clin Transl Immunology* 2021; **10**: e1295.
- 18 Torke S, Pretzsch R, Häusler D, et al. Inhibition of Bruton's tyrosine kinase interferes with pathogenic B-cell development in inflammatory CNS demyelinating disease. *Acta Neuropathol* 2020; **140**: 535–48.
- 19 Li R, Tang H, Burns JC, et al. BTK inhibition limits B-cell-T-cell interaction through modulation of B-cell metabolism: implications for multiple sclerosis therapy. *Acta Neuropathol* 2022; **143**: 505–21.
- 20 Li R, Lei Y, Rezk A, et al, and the Diego A. Espinoza. Oxidative phosphorylation regulates B cell effector cytokines and promotes inflammation in multiple sclerosis. *Sci Immunol* 2024; **9**: eadk0865.
- 21 Menzfeld C, John M, van Rossum D, et al. Tyrphostin AG126 exerts neuroprotection in CNS inflammation by a dual mechanism. *Glia* 2015; **63**: 1083–99.
- 22 Nuesslein-Hildesheim B, Ferrero E, Schmid C, et al. Remibrutinib (LOU064) inhibits neuroinflammation driven by B cells and myeloid cells in preclinical models of multiple sclerosis. *J Neuroinflammation* 2023; **20**: 194.
- 23 Bhargava P, Kim S, Reyes AA, et al. Imaging meningeal inflammation in CNS autoimmunity identifies a therapeutic role for BTK inhibition. *Brain* 2021; **144**: 1396–408.
- 24 Incrocci R, Barse L, Stone A, et al. Epstein-Barr virus latent membrane protein 2A (LMP2A) enhances IL-10 production through the activation of Bruton's tyrosine kinase and STAT3. *Virology* 2017; **500**: 96–102.
- 25 Montalban X, Arnold DL, Weber MS, et al, and the Evobrutinib Phase 2 Study Group. Placebo-controlled trial of an oral BTK inhibitor in multiple sclerosis. *N Engl J Med* 2019; **380**: 2406–17.
- 26 Montalban X, Piasecka-Strzyzewska K, Kuhle J, et al. Efficacy and safety results after >3.5 years of treatment with the Bruton's tyrosine kinase inhibitor evobrutinib in relapsing multiple sclerosis: Long-term follow-up of a Phase II randomised clinical trial with a cerebrospinal fluid sub-study. *Mult Scler* 2024; **30**: 558–70.
- 27 Montalban X, Vermersch P, Arnold DL, et al, and the evolutionRMS investigators. Safety and efficacy of evobrutinib in relapsing multiple sclerosis (evolutionRMS1 and evolutionRMS2): two multicentre, randomised, double-blind, active-controlled, phase 3 trials. *Lancet Neurol* 2024; **23**: 1119–32.
- 28 Reich DS, Arnold DL, Vermersch P, et al, and the Tolebrutinib Phase 2b Study Group. Safety and efficacy of tolebrutinib, an oral brain-penetrant BTK inhibitor, in relapsing multiple sclerosis: a phase 2b, randomised, double-blind, placebo-controlled trial. *Lancet Neurol* 2021; **20**: 729–38.
- 29 Oh J, Fox RJ, Arnold DL, et al. Tolebrutinib phase 2b long-term extension study: two-year safety, MRI, and clinical efficacy outcomes in patients with relapsing multiple sclerosis patients. *Neurol Open Access* 2025; **1**: 3.
- 30 Oh J, Arnold DL, Cree BAC, et al, and the Tolebrutinib Phase 3 GEMINI 1 and 2 Trial Group. Tolebrutinib versus teriflunomide in relapsing multiple sclerosis. *N Engl J Med* 2025; **392**: 1893–904.
- 31 Fox RJ, Bar-Or A, Traboulsee A, et al, and the HERCULES Trial Group. Tolebrutinib in nonrelapsing secondary progressive multiple sclerosis. *N Engl J Med* 2025; **392**: 1883–92.
- 32 Bar-Or A, Dufek M, Budincevic H, et al, and the FENopta Study Group. Safety and efficacy of fenebrutinib in relapsing multiple sclerosis (FENopta): a multicentre, double-blind, randomised, placebo-controlled, phase 2 trial and open-label extension study. *Lancet Neurol* 2025; **24**: 656–66.
- 33 Arnold DL, Elliott C, Martin EC, Hyvert Y, Tomic D, Montalban X. Effect of evobrutinib on slowly expanding lesion volume in relapsing multiple sclerosis: a post hoc analysis of a phase 2 trial. *Neurology* 2024; **102**: e208058.
- 34 Oh J, Reich D, Traboulsee A, et al, eds. Safety and clinical efficacy outcomes from the long-term extension study of tolebrutinib in participants with relapsing multiple sclerosis: 3-year results. Ninth Joint ECTRIMS-ECTRIMS Meeting. Oct 11–13, 2023 (abstr P278).
- 35 Oh J, Drulovic J, Dufek M, et al, eds. Fenebrutinib maintains early and sustained low disease activity in patients with relapsing multiple sclerosis: 2-year results from the FENopta open-label extension. ECTRIMS. Sept 24–26, 2025 (abstr 0112).
- 36 Xu Y, Cui L, Sowa M, et al, eds. Positive phase 2 results of orelabrutinib in patients with relapsing-remitting multiple sclerosis. ACTRIMS. Feb 27 to March 1, 2025 (abstr P094).
- 37 Oh J, Cree B, Arnold D, et al, eds. Effects of tolebrutinib on progression independent of relapse activity in the phase 3 GEMINI relapsing MS trials ECTRIMS. Sept 24–26, 2025 (abstr P794).
- 38 Oh J, Fox RJ, Arnold DL, et al. Paramagnetic rim lesions as a prognostic and predictive biomarker in the tolebrutinib phase 3 trials for disability outcomes. ACTRIMS. Feb 27 to March 1, 2025 (abstr LB1.1).
- 39 Roche's fenebrutinib shows unprecedented positive Phase III results as the potential first and only BTK inhibitor in both relapsing and primary progressive multiple sclerosis 2025. <https://www.roche.com/investors/updates/inv-update-2025-11-10> (accessed April 19, 2026).
- 40 Bar-Or A, Oh J, Giovannoni G, et al. Efficacy and safety of fenebrutinib vs ocrelizumab in primary progressive multiple sclerosis: primary results of the phase III FENTrepid study. ACTRIMS. Feb 7, 2026 (abstr 004).
- 41 Reich D, Bar-Or A, Vermersch P, et al. Efficacy and safety of tolebrutinib versus placebo in primary progressive multiple sclerosis: results from the phase 3 PERSEUS trial. ACTRIMS. Feb 7, 2026 (abstr LB1.4).
- 42 Owens TD, Smith PF, Redfern A, et al. Phase 1 clinical trial evaluating safety, exposure and pharmacodynamics of BTK inhibitor tolebrutinib (PRN2246, SAR442168). *Clin Transl Sci* 2022; **15**: 442–50.
- 43 Caldwell RD, Qiu H, Askew BC, et al. Discovery of evobrutinib: an oral, potent, and highly selective, covalent Bruton's tyrosine kinase (BTK) inhibitor for the treatment of immunological diseases. *J Med Chem* 2019; **62**: 7643–55.
- 44 Cabanis M, Nicolas O, Vitse O, et al. A phase I trial assessing the safety, pharmacokinetics, cerebrospinal fluid penetrance, and food effect of BTK inhibitor tolebrutinib in healthy volunteers. *Clin Transl Sci* 2024; **17**: e13693.
- 45 Crawford JJ, Johnson AR, Misner DL, et al. Discovery of GDC-0853: a potent, selective, and noncovalent Bruton's tyrosine kinase inhibitor in early clinical development. *J Med Chem* 2018; **61**: 2227–45.
- 46 Turner TJ, Brun P, Gruber RC, Ofengeim D. Comparative CNS pharmacology of the Bruton's tyrosine kinase (BTK) inhibitor tolebrutinib versus other BTK inhibitor candidates for treating multiple sclerosis. *Drugs R D* 2024; **24**: 263–74.
- 47 Herman AE, Chinn LW, Kotwal SG, et al. Safety, pharmacokinetics, and pharmacodynamics in healthy volunteers treated with GDC-0853, a selective reversible Bruton's tyrosine kinase inhibitor. *Clin Pharmacol Ther* 2018; **103**: 1020–28.
- 48 Weber M, Harp C, Bremer M, et al, eds. Fenebrutinib demonstrates the highest potency of Bruton tyrosine kinase inhibitors (BTKis) in phase 3 clinical development for multiple sclerosis (MS). AAN. April 17–22, 2021 (abstr 4437).
- 49 Lefèvre G, Tian X, Choudhury S, et al. Remibrutinib exposure in cerebrospinal fluid: insights from a study in healthy subjects. *Neurology* 2025; **104**: 1212.
- 50 Estupiñán HY, Berglöv A, Zain R, Smith CIE. Comparative analysis of BTK inhibitors and mechanisms underlying adverse effects. *Front Cell Dev Biol* 2021; **9**: 630942.
- 51 Miklos D, Cutler CS, Arora M, et al. Ibrutinib for chronic graft-versus-host disease after failure of prior therapy. *Blood* 2017; **130**: 2243–50.
- 52 Lipsky A, Lamanna N. Managing toxicities of Bruton tyrosine kinase inhibitors. *Hematology* 2020; **2020**: 336–45.
- 53 Ringheim GE, Wampole M, Oberoi K. Bruton's tyrosine kinase (BTK) inhibitors and autoimmune diseases: making sense of BTK inhibitor specificity profiles and recent clinical trial successes and failures. *Front Immunol* 2021; **12**: 662223.

- 54 Pulz R, Angst D, Cenni B. Next generation Bruton's tyrosine kinase inhibitors: characterization of in vitro potency and selectivity. *Eur J Pharmacol* 2025; **1002**: 177747.
- 55 Himmelbauer MK, Bajrami B, Basile R, et al. Discovery and preclinical characterization of BIIB129, a covalent, selective, and brain-penetrant BTK inhibitor for the treatment of multiple sclerosis. *J Med Chem* 2024; **67**: 8122–40.
- 56 Haselmayer P, Camps M, Liu-Bujalski L, et al. Efficacy and pharmacodynamic modeling of the BTK inhibitor evobrutinib in autoimmune disease models. *J Immunol* 2019; **202**: 2888–906.
- 57 Hopkins BT, Bame E, Bajrami B, et al. Discovery and preclinical characterization of BIIB091, a reversible, selective BTK inhibitor for the treatment of multiple sclerosis. *J Med Chem* 2022; **65**: 1206–24.
- 58 Brullo C, Villa C, Tasso B, Russo E, Spallarossa A. Btk inhibitors: a medicinal chemistry and drug delivery perspective. *Int J Mol Sci* 2021; **22**: 7641.
- 59 Zain R, Vihinen M. Structure-function relationships of covalent and non-covalent BTK inhibitors. *Front Immunol* 2021; **12**: 694853.
- 60 Gaitan MI, Raza S, Donnay C, et al, eds. Primary outcome of a phase 2 clinical trial of tolebrutinib, a brain-penetrant BTK inhibitor, for the modulation of chronically inflamed white matter lesions in MS. ACTRIMS. Feb 29 to March 2, 2024 (abstr LB1.2).
- 61 Blazier AS, Wirak G, Krishnaswami P, et al, eds. Tolebrutinib can reverse multiple sclerosis-induced cerebrospinal fluid proteomic alterations. ACTRIMS. Feb 23–25, 2023 (abstr P019).
- 62 Tafesh ZH, Coleman M, Fulmer C, Nagler J. Severe hepatotoxicity due to ibrutinib with a review of published cases. *Case Rep Gastroenterol* 2019; **13**: 357–63.
- 63 Kleijwegt FS, Roda AA, Rolvink J, Kater AP, Kersten MJ, Vos JMI. Rare but serious: ibrutinib induced liver failure. *HemaSphere* 2019; **3**: e307.
- 64 Pleyer C, Ali MA, Cohen JI, et al. Effect of Bruton tyrosine kinase 64 on efficacy of adjuvanted recombinant hepatitis B and zoster vaccines. *Blood* 2021; **137**: 185–89.
- 65 Herzog Tzarfati K, Gutwein O, Apel A, et al. BNT162b2 COVID-19 vaccine is significantly less effective in patients with hematologic malignancies. *Am J Hematol* 2021; **96**: 1195–203.
- 66 FDA. Complete response letter for tolebrutinib NDA. 2025. https://download.open.fda.gov/crl/CRL_NDA219624_20251223.pdf (accessed April 9, 2026).
- 67 Sanofi. Press release: Sanofi provides update on tolebrutinib regulatory submission in non-relapsing secondary progressive multiple sclerosis. 2025. <https://www.sanofi.com/en/media-room/press-releases/2025/2025-12-24-06-00-00-3210238> (accessed April 19, 2026).

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